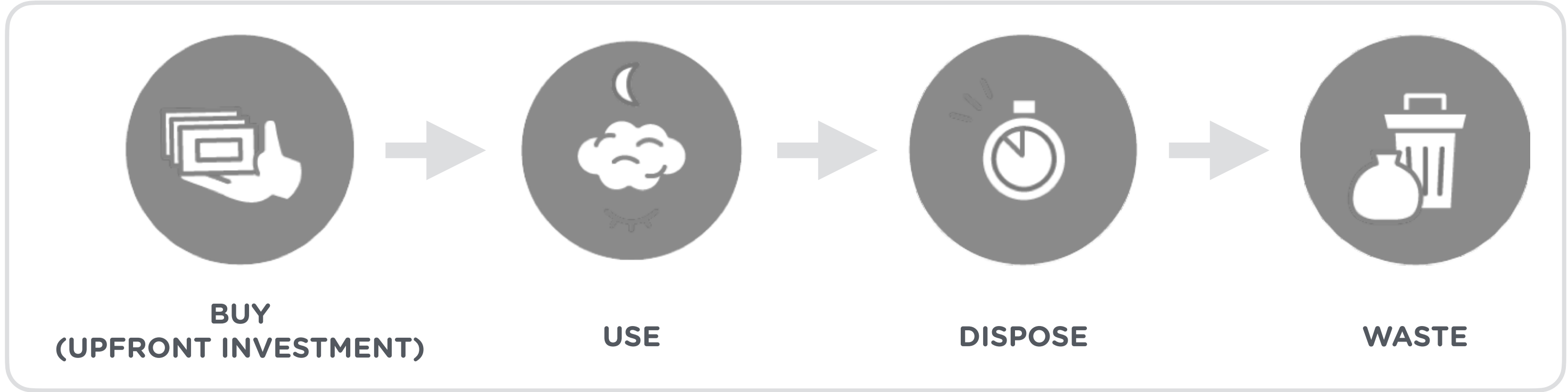


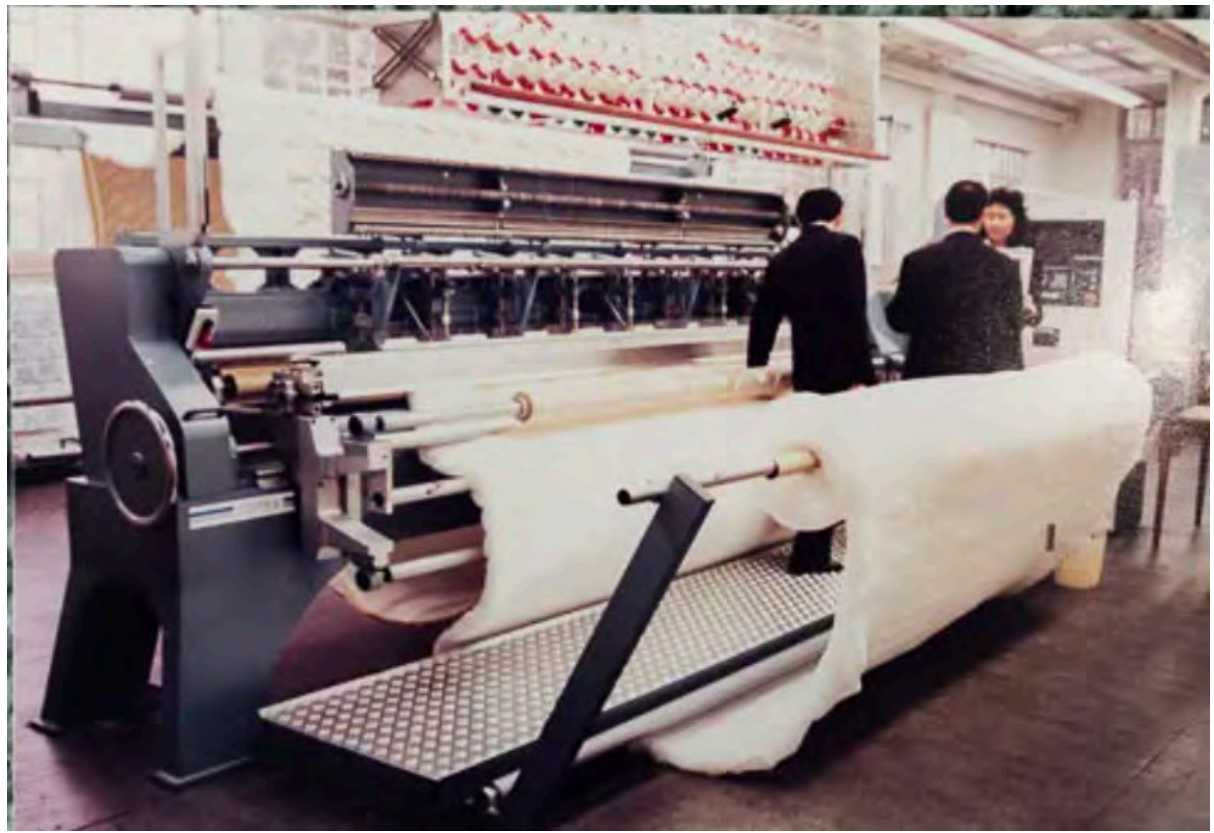


circular economy product subscription platform



OUR VISION: A GREENER, CIRCULAR ECONOMY

OUR FOUNDER'S 95-YEAR HERITAGE IN THE MATTRESS INDUSTRY



ENTREPRENEURS/THAILAND



Springmate:

เมื่อโอกาสกำลังเคาะประตู

โชคและโอกาสเป็นองค์ประกอบแห่งความสำเร็จที่นักธุรกิจจำนวนมากทวิหา แต่มีนักธุรกิจเพียงไม่กี่คนที่สามารถไขคว้ามวยอดได้ดังฝัน เพราะความสำเร็จไม่ได้เกิดขึ้นเพียงเพราะโชคและโอกาส

เรื่อง: พรพรรณ ปิณฑากิธมย์ ภาพ: สุทธิ สุขปลั่ง

Joseph Philippe Pierre Yves Elliott Trudeau นายกรัฐมนตรีคนที่ 15 ของแคนาดา เคยกล่าวไว้ว่า Be ready when opportunity comes... Luck is the time when preparation and opportunity meet. ประโยคที่สามารถอธิบายภาพความสำเร็จในวันนี้ของธุรกิจ

เครื่องนอนแบรนด์ไทย ภายใต้ชื่อ สปริงเมท (Springmate) ของทายาทรุ่น 2 นพพล เดชะพันธ์งาม รองกรรมการผู้จัดการ บริษัท สมพลเบตติง แอนด์ แมทเทรส อินดัสตรี จำกัด (SPB) ได้เป็นอย่างดี เพราะทุกอย่างก้าวทางธุรกิจเกิดขึ้นบนความเชื่อที่ว่า “เราต้องพยายามทำตัวเองให้พร้อมทุกด้าน เพราะโชคจะช่วยคนที่พร้อมกว่าเสมอ”

ย้อนกลับไปเมื่อปี 2534 นพพลในวัย 9 ปี มองเห็นธุรกิจของครอบครัวเริ่มต้นขึ้นอย่างแตกต่างจากผู้ประกอบการรายอื่น ขณะที่คู่แข่งเน้นการสร้างแบรนด์ ทว่าสปริงเมทเลือกเป็นผู้รับจ้างผลิตที่นอนและเครื่องนอนที่เน้นคุณภาพสูงในภูมิภาคเอเชียตะวันออกเฉียงใต้ และเป็นผู้ให้บริการรับสั่งทำที่นอน และเครื่องนอนที่ได้รับความนิยมไว้

84 | FORBES THAILAND JULY 2013

July 2013

THAILAND
Forbes



Unaffordability of
quality mattresses



Unethical and
eco-damaging disposal



No mattress recycling

Estimated global number of mattresses disposed of annually:

> 150 million*

*Reference: "End of Life Mattress Report 2019" by the National Bed Federation (UK); Oakdene Hollins, July 2019



Environment ► Climate change Wildlife More



Waste

The mattress landfill crisis: how the race to bring us better beds led to a recycling nightmare

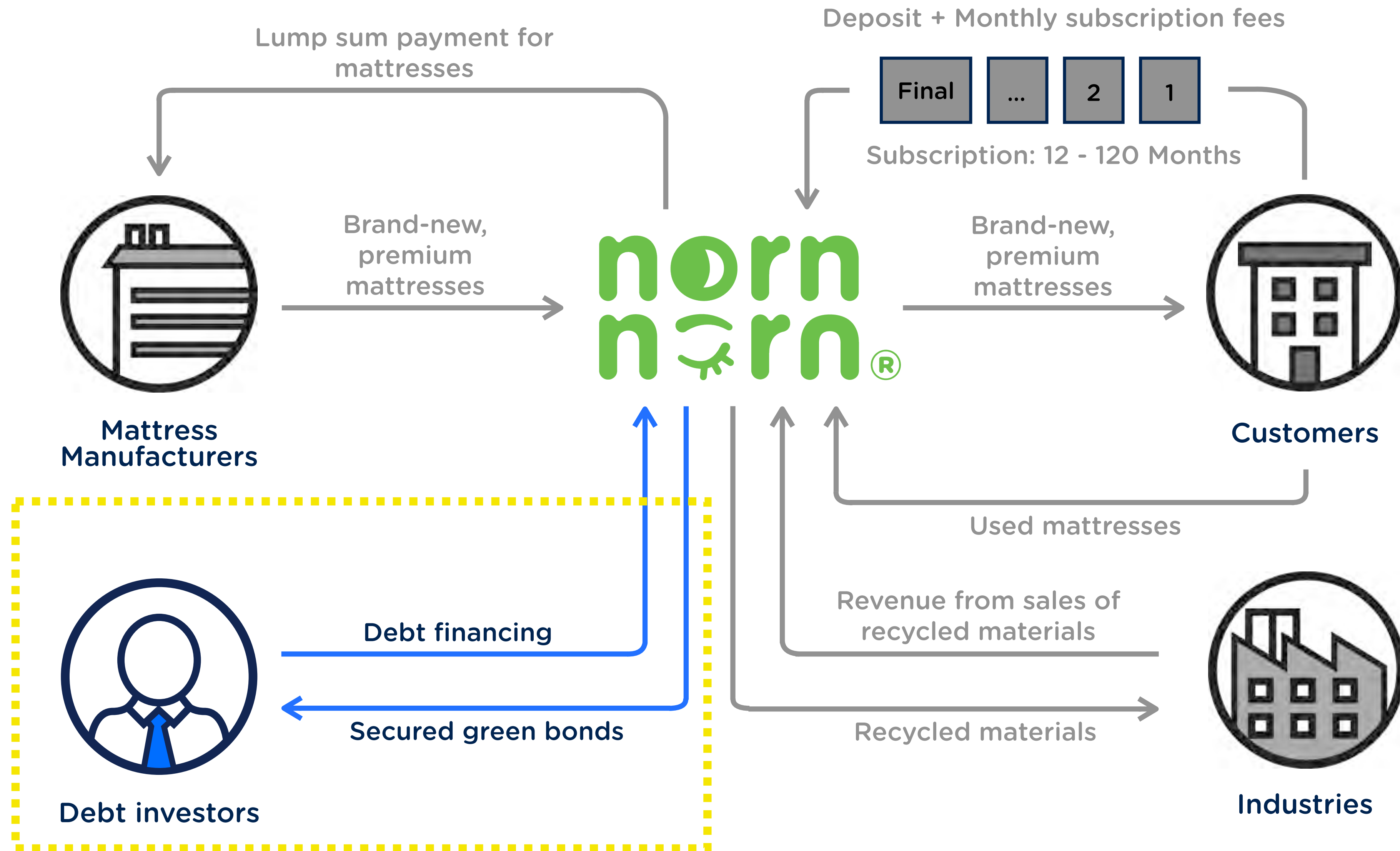


WORLD'S FIRST CIRCULAR ECONOMY MATTRESS SUBSCRIPTION PLATFORM

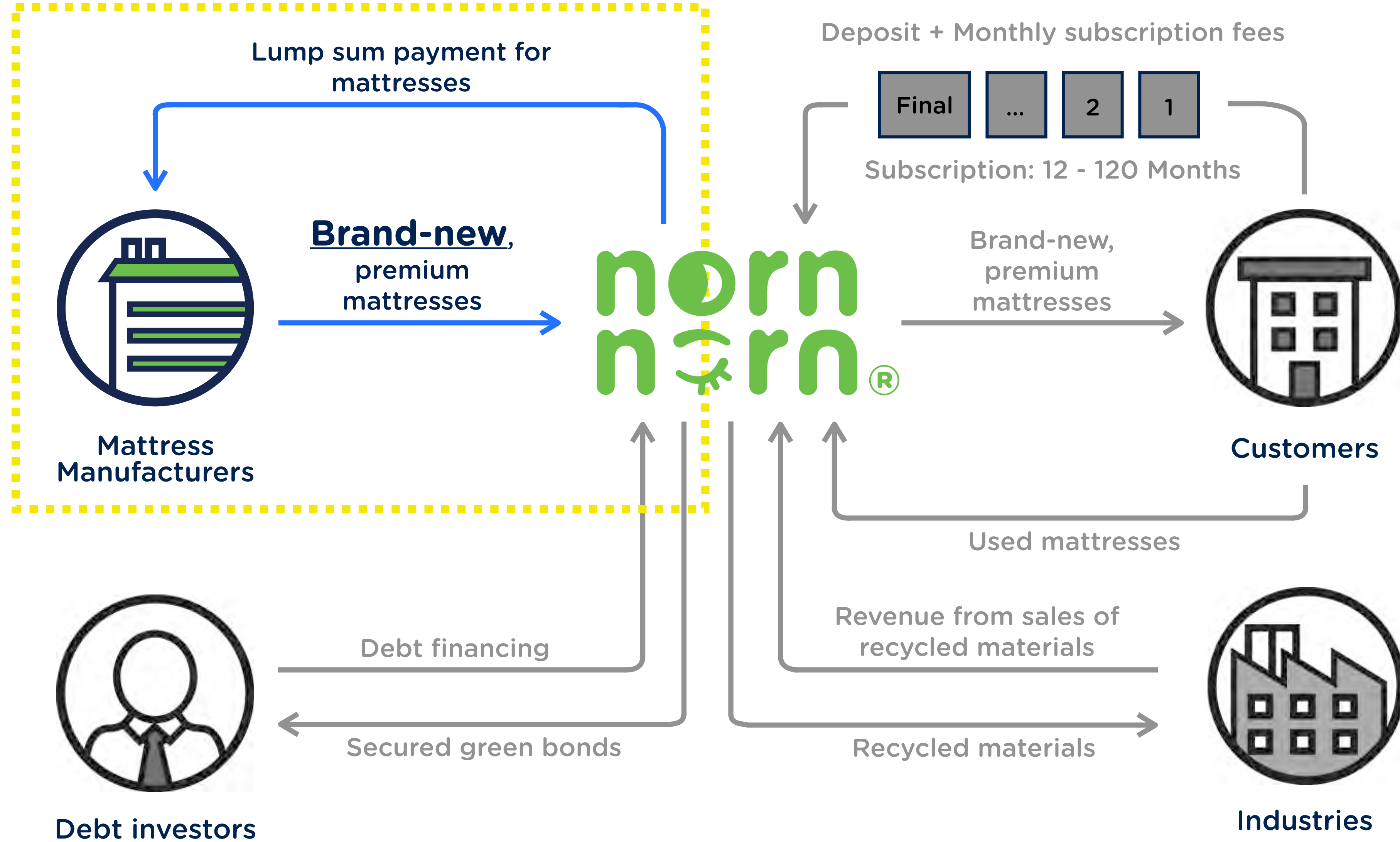




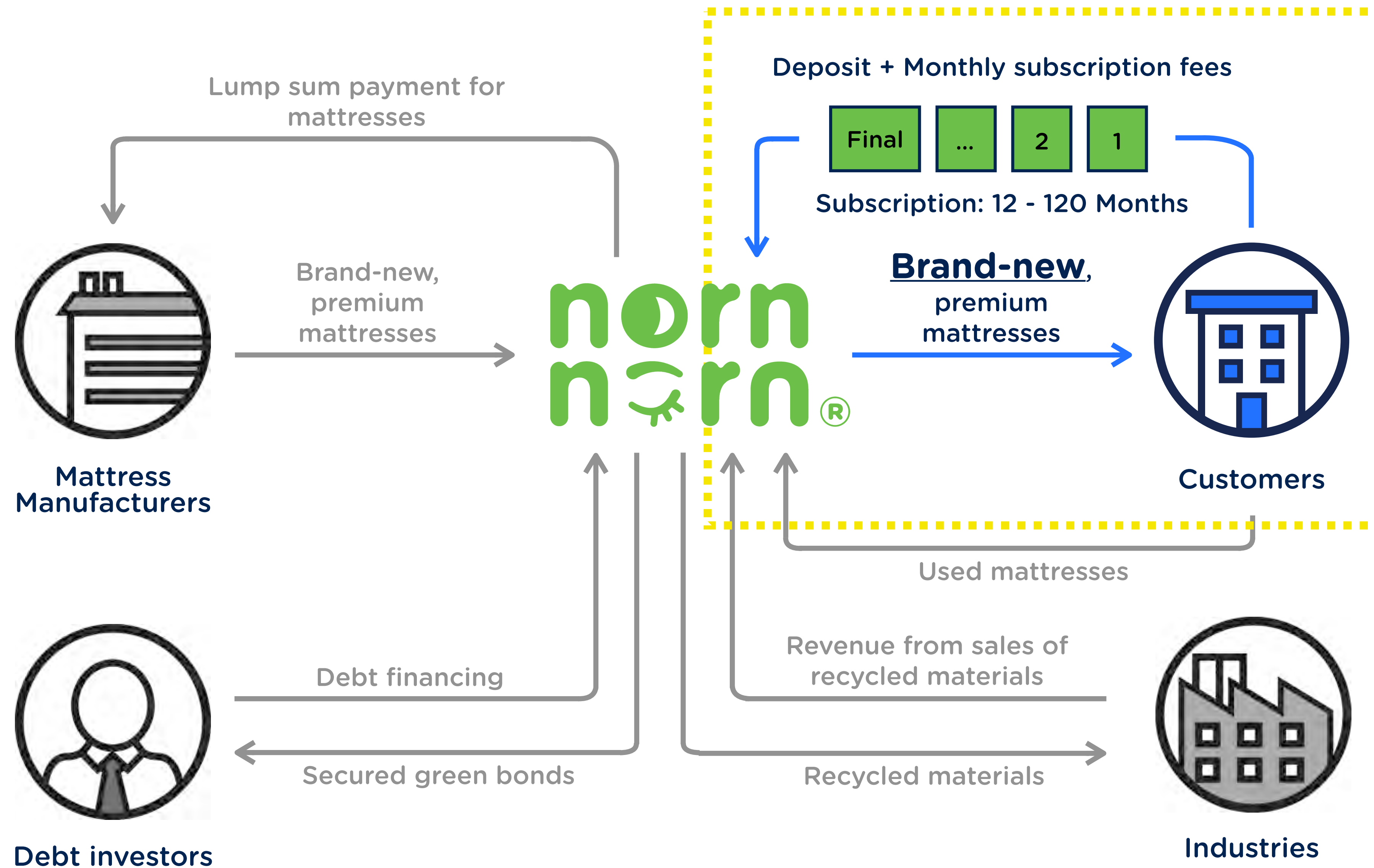
STEP 1: DEBT FINANCING



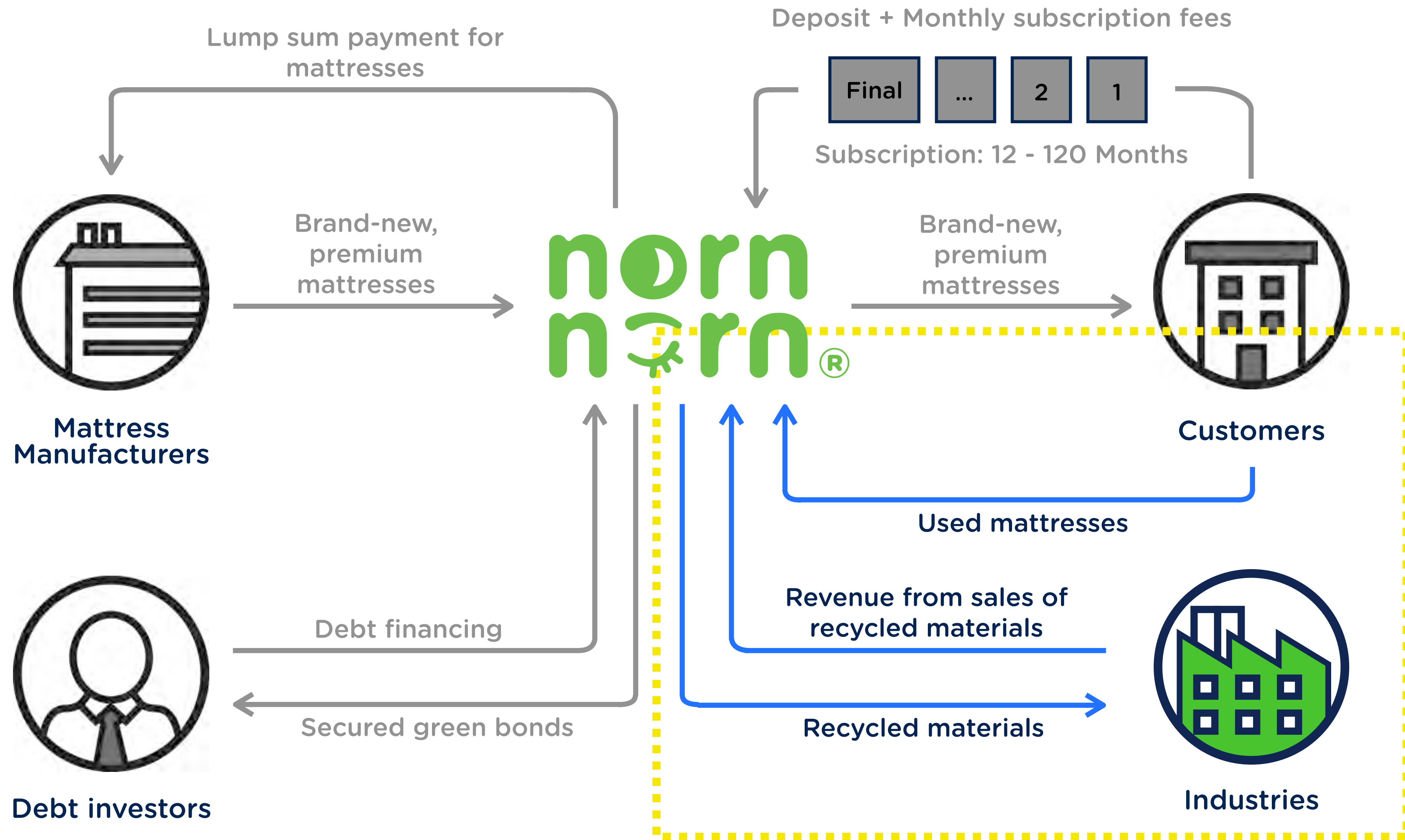
STEP 2: PRODUCT ACQUISITION



STEP 3: FULFILLMENT



STEP 4: PRODUCT RETRIEVAL & RECYCLING



OUR PREMIUM MATTRESSES

From THB89/mattress/month



สปริงเมท: Project Superior
เริ่มต้น THB112.00 บาท/เดือน



สปริงเมท: Project Premier
เริ่มต้น THB135.00 บาท/เดือน



สปริงเมท: Pocket Coil Heritage
เริ่มต้น THB151.00 บาท/เดือน



สปริงเมท: Pocket Coil Heritage PT
เริ่มต้น THB157.00 บาท/เดือน

duroflex

South Asia



Indonesia



Thailand

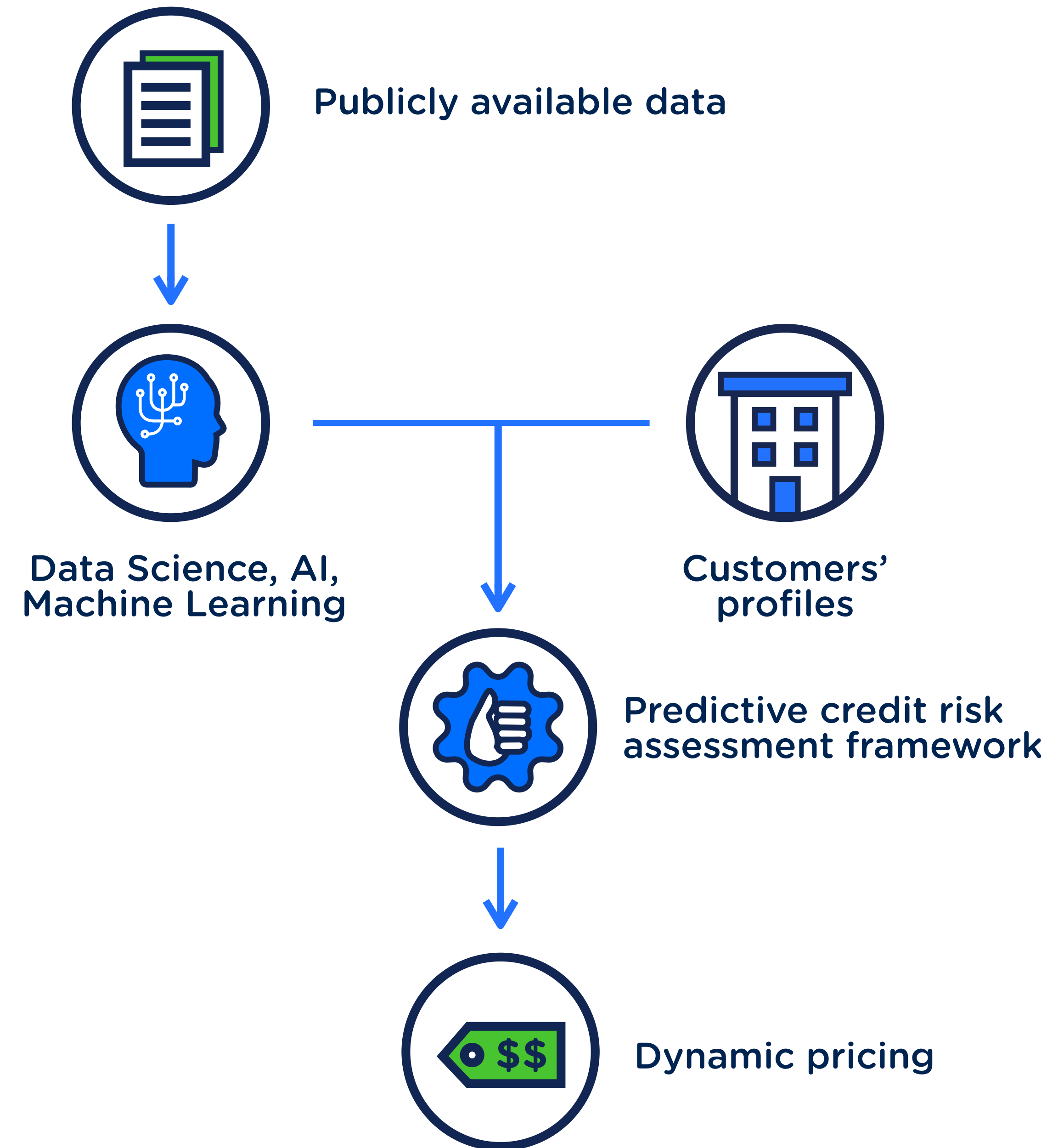
OUR TECHNOLOGY STACK*

1) Non-intrusive predictive credit risk assessment framework powered by:

- Publicly available historical industry and economic data
- Data Science, AI and Machine Learning
- Customers' profiles

2) Dynamic pricing based on customers' credit risk profiles

3) Financially sustainable and eco-friendly recycling technologies



*Under development

WHY SUBSCRIBE?

1. Affordable access to premium mattresses.
2. Support end-of-life product recycling.

“We strive to be sustainable and conscious in every way possible, so when we learned about nornnorn’s offering while figuring out our procurement, it was a no brainer. Guests are not only inspired by the circular concept, but also find the mattresses super comfortable. The elimination of upfront investment on mattresses is also helpful.”

Tomohiro Hamakawa
Co-Founder, Mana Earthly Paradise, Bali



WHY WE ARE RELEVANT

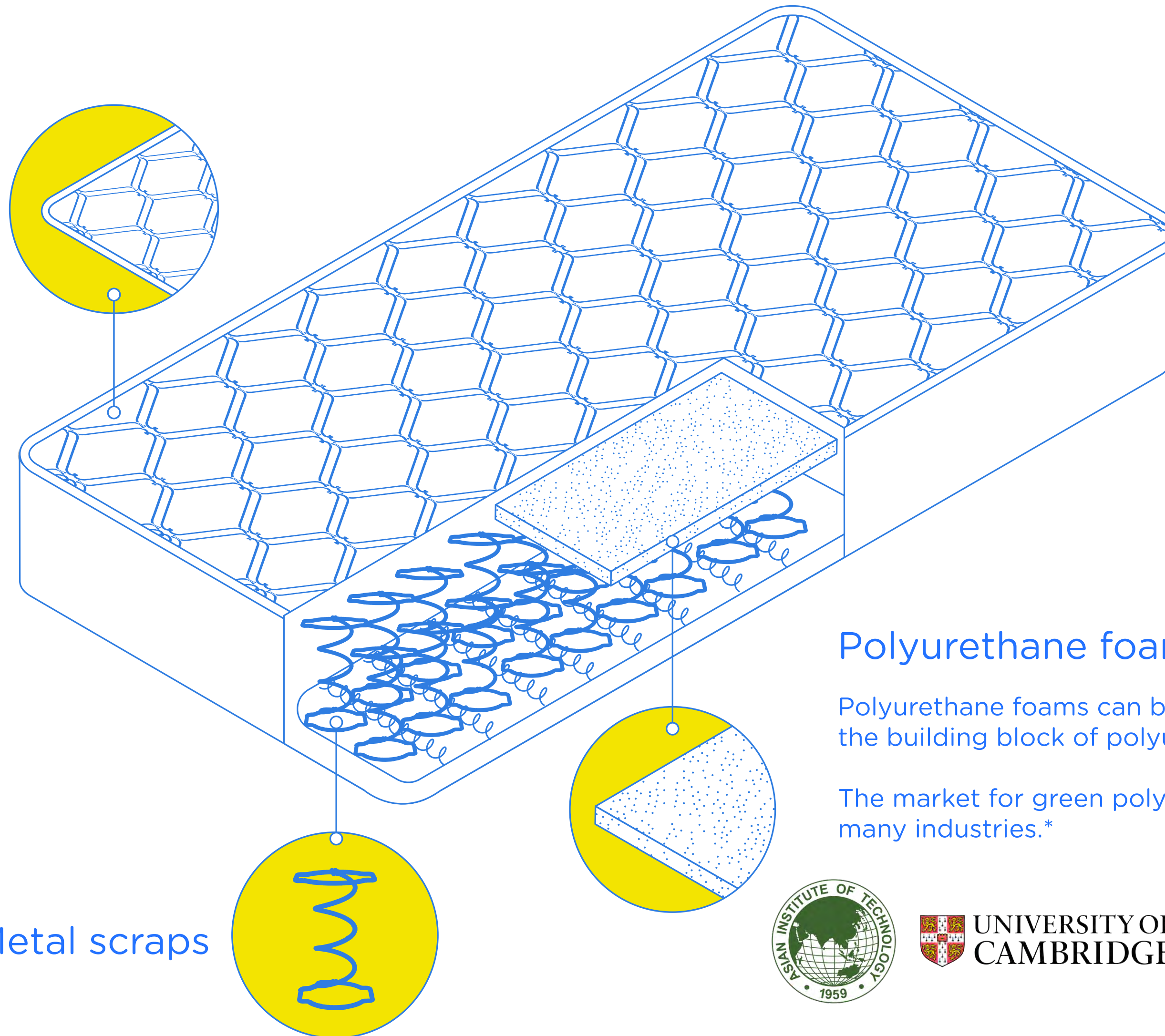
- 1) Rapidly growing demand for subscription-based access to premium products.
- 2) Growing demand for affordable eco-friendly business solutions driven by worsening environmental crises.
- 3) Growing demand for recycled materials due to virgin material shortage.
- 4) Circular-economy business model like ours is the world's future operating system.



From left: Al Jazeera, April 21, 2021; UN Climate Change, News April 15, 2021

CREATING VALUE OUT OF DISPOSED MATTRESSES

Fabrics and fibres →
Industrial oil filters, etc.



Steel springs → Metal scraps

Polyurethane foams → Polyol monomers

Polyurethane foams can be chemically broken down into polyol monomers, the building block of polyurethane foams and a wide range of products.

The market for green polyols is expected to grow due to high demand from many industries.*

Used mattresses will be retrieved from customers by nornnorn free of charge and disassembled at nornnorn-operated materials recovery facilities (MRFs).

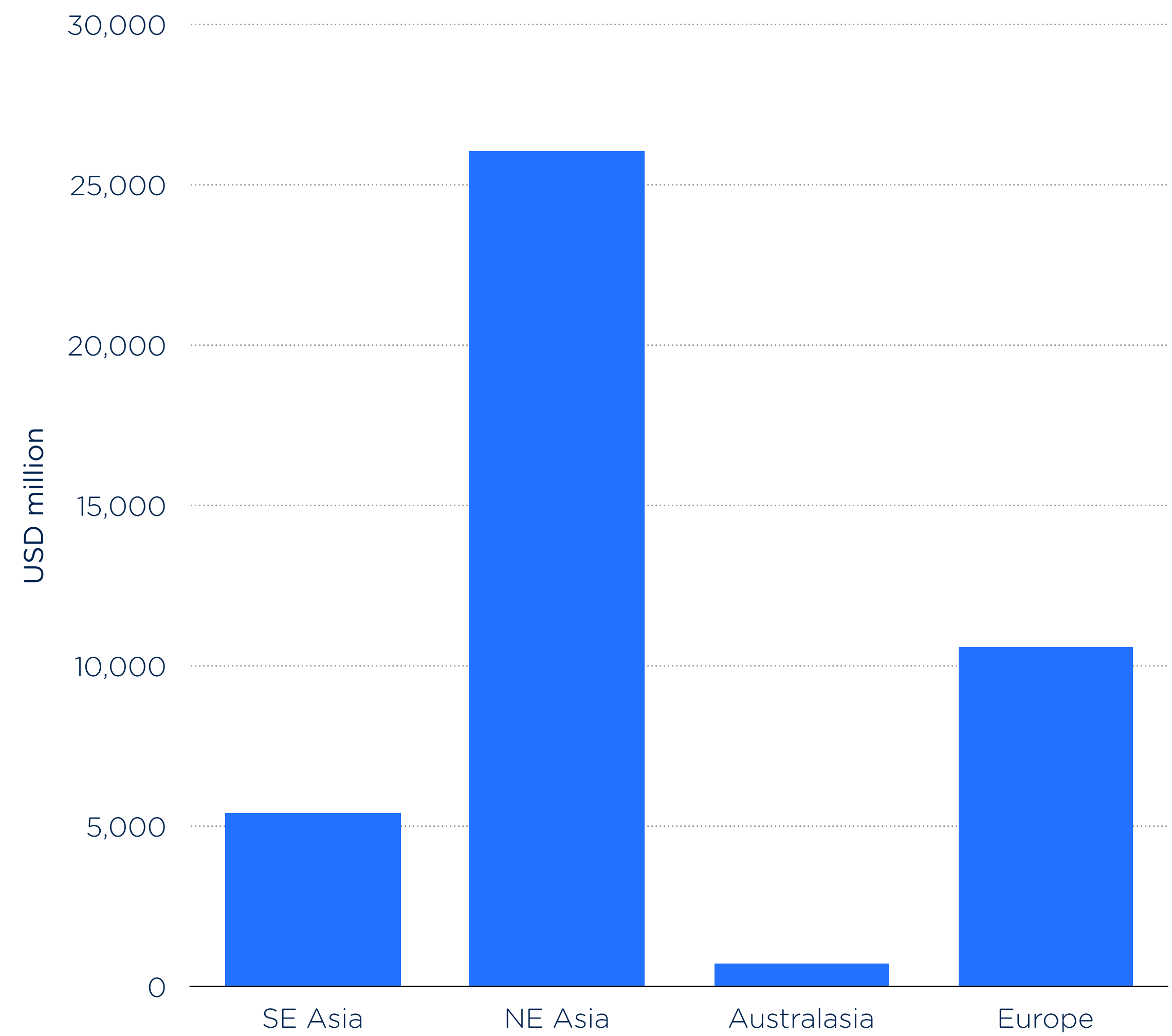
The recovered materials will then be recycled/upcycled either at nornnorn-operated facilities or sold to third parties.

Details on the upcycling technologies we are currently developing are on Slide 21.



*Source: <https://www.marketsandmarkets.com/Market-Reports/polyols-market-725.html/>

ADDRESSABLE MARKET SIZE: B2C

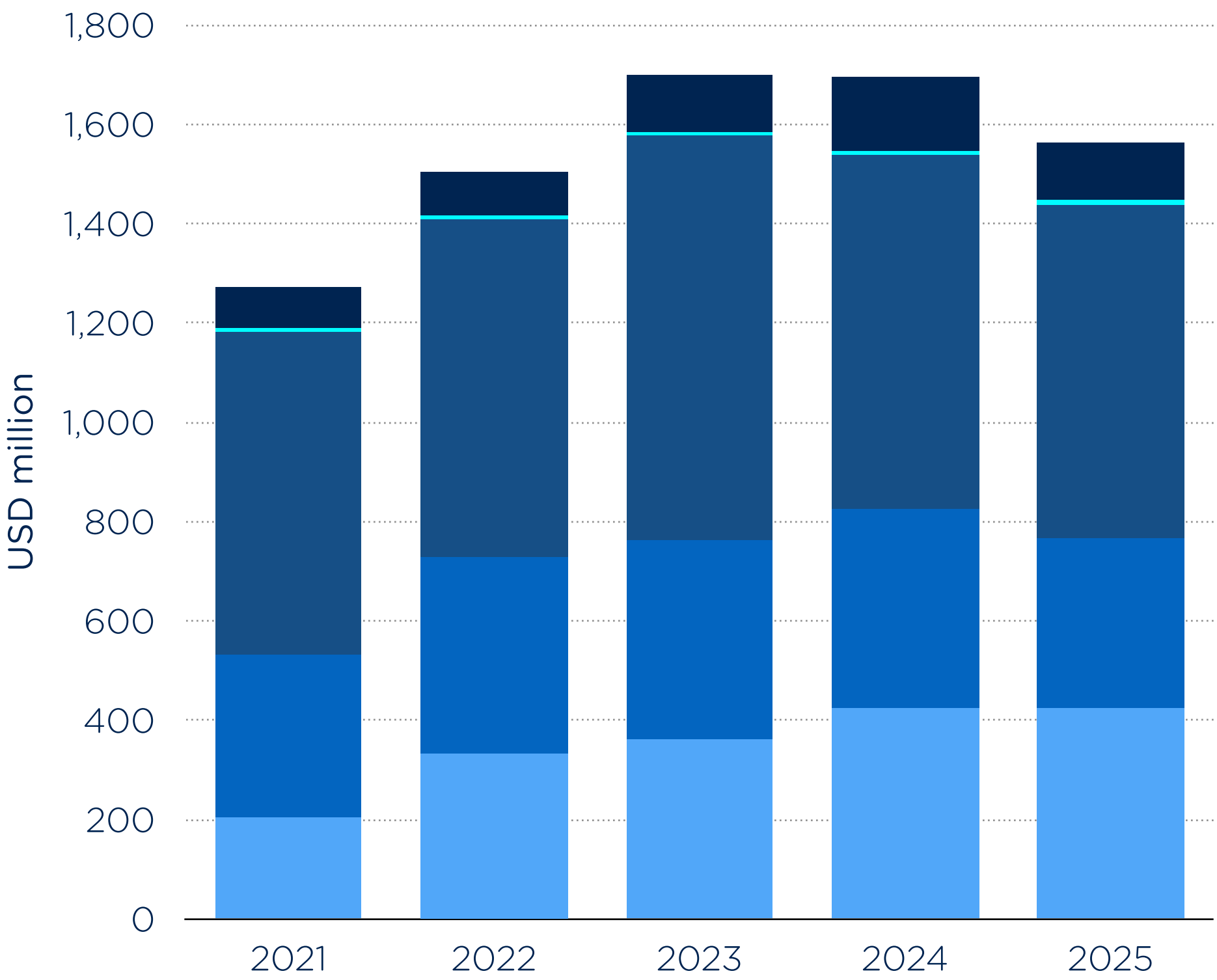


In 2023, the annual addressable B2C market size in SE Asia, our beachhead market, > **US\$5 billion**, with the global market size > **US\$80 billion**.

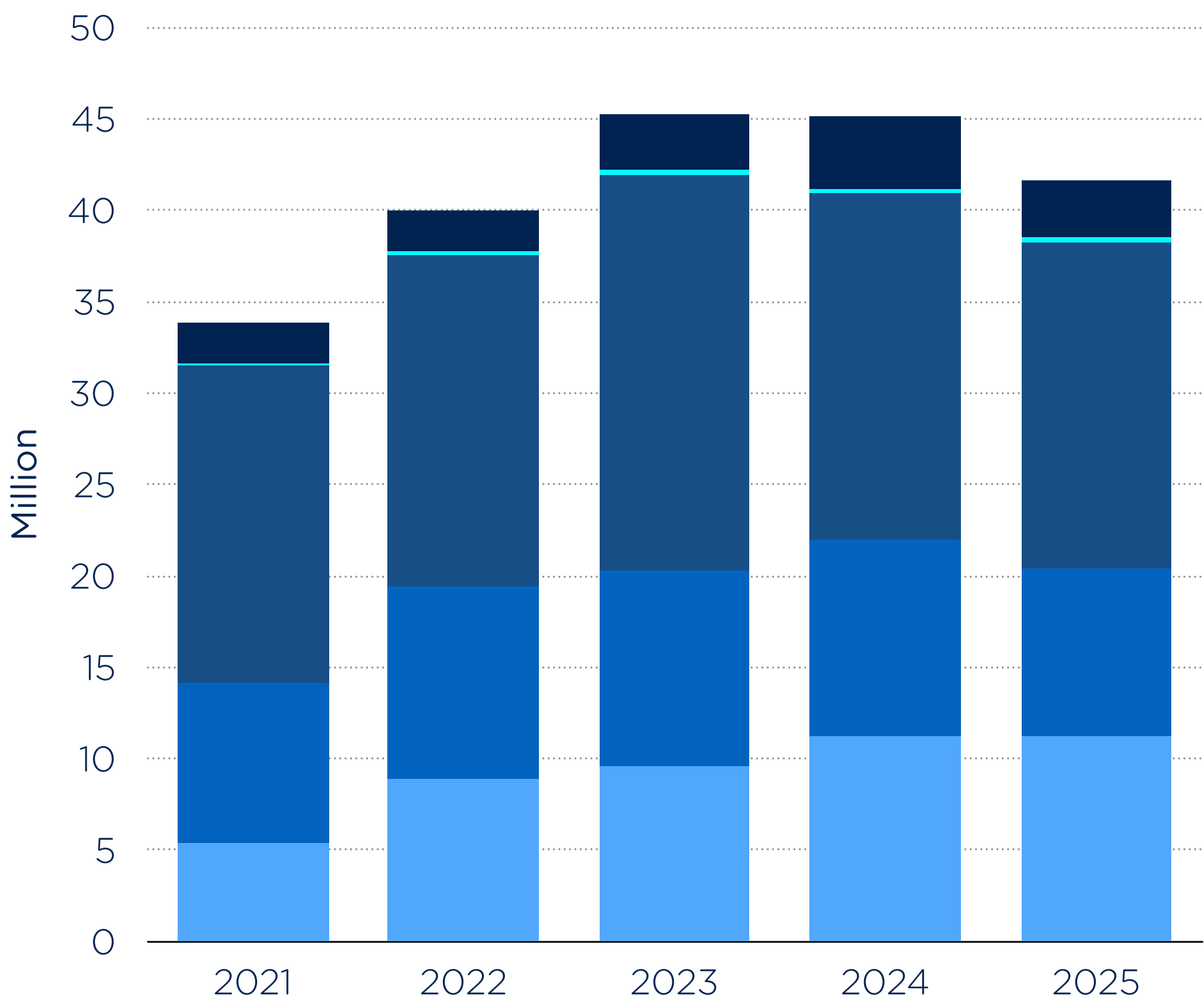
Note: 1) Average monthly subscription fee = USD8.55/mattress/month for 60 months. 2) One mattress per 2 working adults with monthly income > THB25k each. 3) Data sources available upon request. 4) Southeast Asia includes Thailand, Indonesia, Malaysia, Vietnam, the Philippines and Singapore. 2) Northeast Asia includes Taiwan, South Korea, Japan and China. 3) Australasia includes Australia and New Zealand. 4) Europe includes France, Spain, Sweden, Norway, Germany, Finland, Poland, Italy, UK, Romania, Greece, Hungary, Portugal, Austria, Ireland, Denmark, Switzerland, the Netherlands and Turkey. 4) Income distribution: <https://wid.world/> 5) No. of working adults: <https://tradingeconomics.com/>

SE Asia South Asia NE Asia Australasia Middle East

Addressable Market Size



Number of New Mattresses Needed

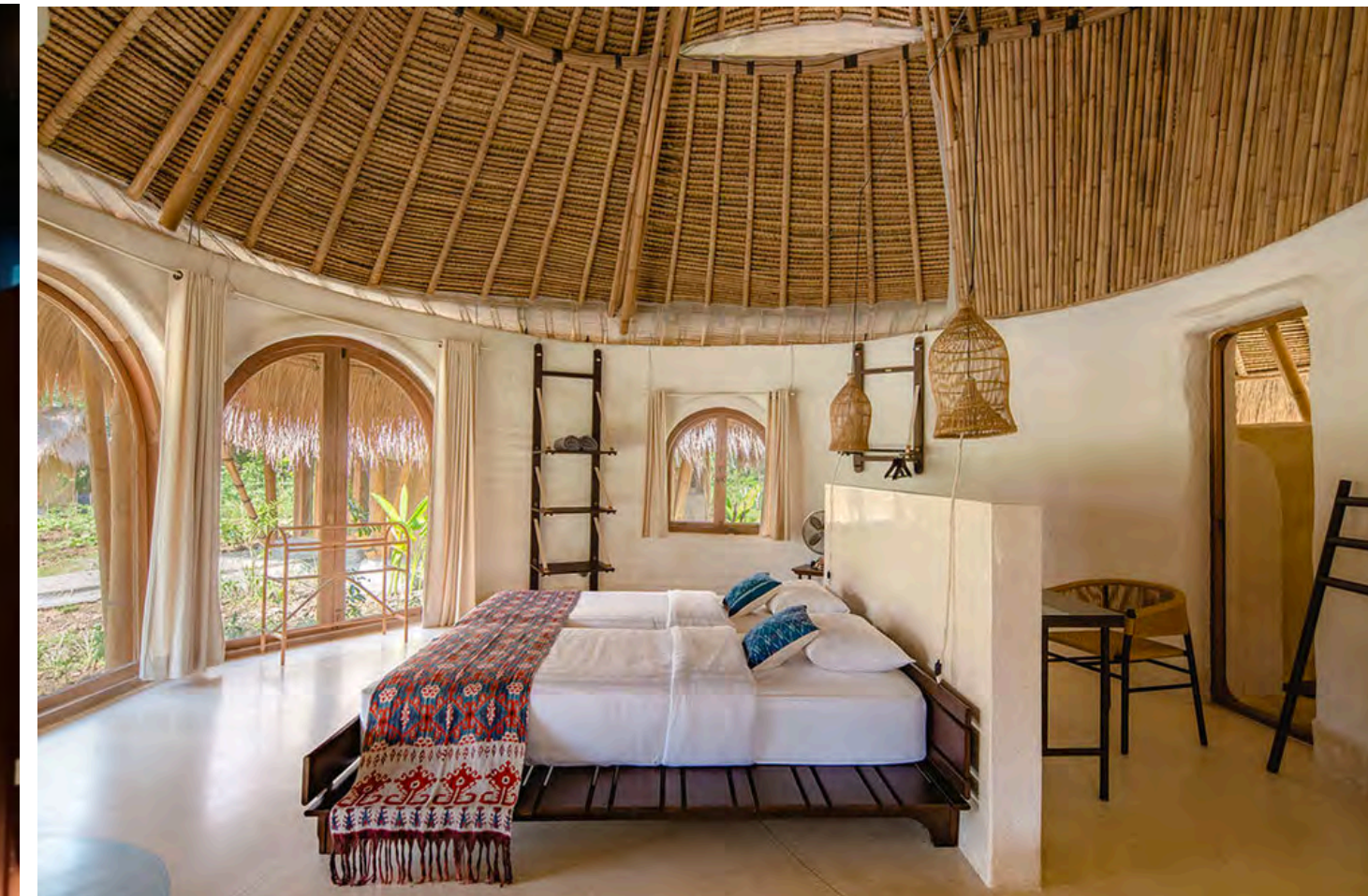


In 2023, the annual addressable B2B market size in SE Asia, our beachhead market, > US\$359M, with the global market size > US\$2.8 billion.

Note: 1) Average monthly subscription fee = USD6.26/room/month for 84 months. 2) Each hotel room requires two single-size mattresses. 3) Data sources available upon request. 4) Southeast Asia excludes Brunei. 5) South Asia excludes Pakistan, Bhutan, Bangladesh and Afghanistan. 6) Middle East excludes Akrotiri and Dhekelia, Iraq, Lebanon, Palestine, Syria and Yemen. 7) East Asia excludes North Korea. 8) Hotels account for only 15% of the total accommodation market. 9) Asia and Australasia account for 35% of the world’s travel and tourism; Source: <https://wtctweb.on.uat.co/Research/Economic-Impact>

Operational since Q3 2018.

460 mattresses* have so far been deployed:



Creative Migration



บ้านพักประวัติศาสตร์
หลวงราชไมตรี
Baan Luang Rajamaitri Historic Inn.

AMANPURI

*In standard size equivalence (107 x 198 x 24 cm.)

OUR FIRST GREEN BOND

We are the first impact startup in Thailand to issue green bonds.

Issuer	Circularity Co., Ltd., Thailand
Type of Instrument	Thai baht senior and unsecured green debentures
Use of Proceeds	Recycling research and development and rental product acquisition
Company & Issue Rating	Non-rated
Size	THB6.05M (~USD200k)
Placement	Private Placement 10 (PP-10)
Tenor	7 years
Issue Date	December 20, 2021
Coupon	7.94% per annum
Principal Repayment	Bullet payment
Green Bond Standard	ICMA Green Bond Principles
Intended Environmental Impact	The components of all the products rented out will be recycled and/or upcycled at the end of every subscription, leading to a reduction in greenhouse gas emissions, pollutions and wastes.
Special Offer	Call options upon maturity granting the investors the option to purchase shares of the Thailand-registered Circularity Co., Ltd. worth up to 50% of the value of the bonds purchased at the price which equates to the value of the company as of October 9, 2021.

nornnorn’s first green bond investors



As we scale up, our green bonds will be secured with our revenue streams.

OUR RESEARCH & DEVELOPMENT

USD228k worth of grants received from the Thai and British governments.

Project	Objective	Researcher / Developer	Period	Expected Outcome	Funder / Supporter	Project Value (USD)	Support (USD)
Life Cycle Assessment	To estimate the amount of greenhouse gas emissions and wastes our business will save over the coming years.	 Technology and Informatics Institute for Sustainability, Thailand	Apr '21 - Feb '23	Thailand's mattress industry's first-ever LCA report highlighting the environmental benefit of our business model	 Office of National Higher Education Science Research and Innovation Policy Council	34k	28k grant
Polyurethane Foam Upcycling Technology	To maximise the economic value of polyurethane foam from used mattresses through advanced chemical upcycling.	  UNIVERSITY OF CAMBRIDGE	Apr '21 - Dec '23	TRL4 polyurethane foam upcycling technology		361k	100k grant
Polyurethane Foam Upcycling Process Water Extraction Technology	To extract and purify water from polyurethane foam upcycling process.	  UNIVERSITY OF CAMBRIDGE	Apr '21 - Dec '23	TRL4 polyurethane foam upcycling process water extraction technology		100k	100k grant

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OUR RESEARCH & DEVELOPMENT

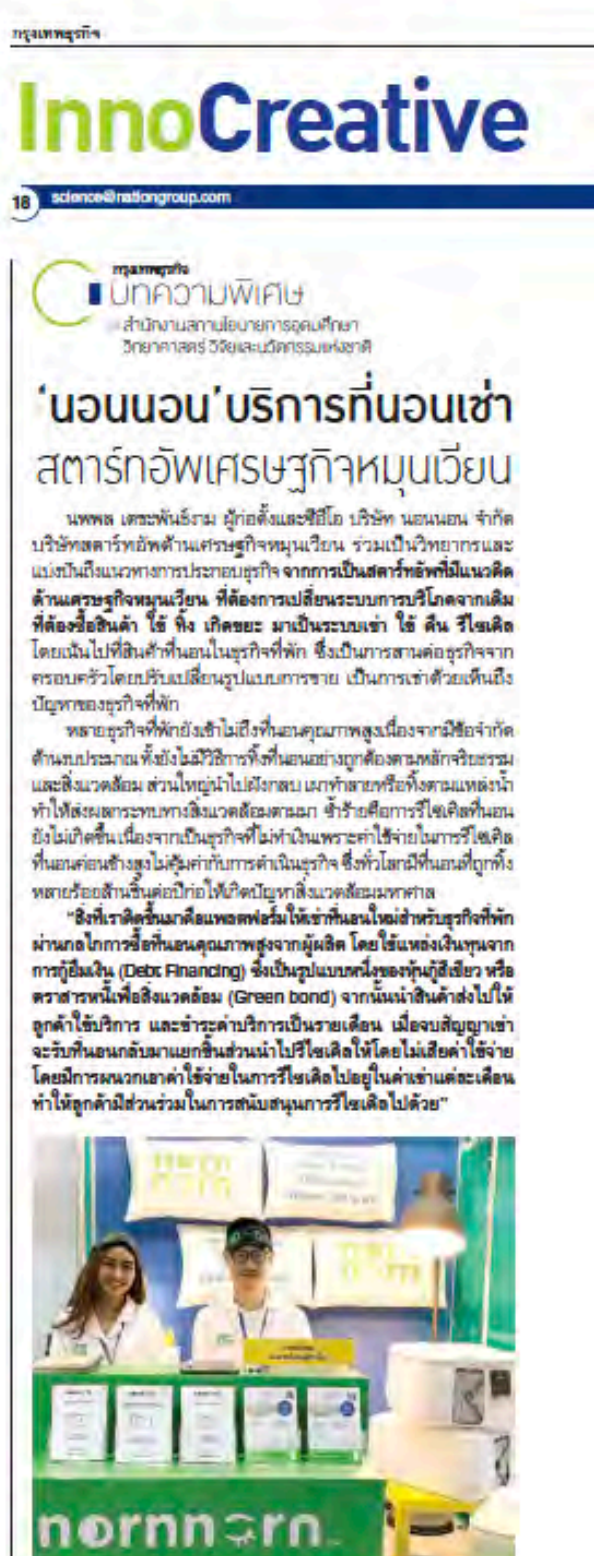
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Top to bottom, from left to right: The Cloud, June 2020; Technode, December 2020; The Futurist, May - July 2021; K SME Inspired Magazine, December 2020; Thailand Tatler, February 2021; ASEANFocus, March 2021





Top to bottom, from left to right: TNN Startup, January 2021; GURU Talk, February 2021; BTimes, October 2020; Bangkok Biz News, August 2021; Bangkok Biz News, August 2021; “Puen Koo Kid”, October 2021; Smart SMEs, FM 106 MHz, October 2021; “Idea Tum Ngern”, Mono 29, October 2021



- Participant, UN Climate Neutral Now (since 2020; “Bronze” Level, 2022)
- One of Göteborg&Co’s “101 Ideas for Sustainable Tourism”, Sweden, November 2020
- Winner (Impacting Category), World Tourism Forum Lucerne Start-up Innovation Camp 2021, Switzerland
- Winner (Model Business, Redesigning Relationships, Green Hospitality and Health & Wellness Impact Categories), crQlr Awards 2021
- Rated as one of top 5 global zero waste travel solutions by StartUs, April 2022
- Participant, Circular Economy Accelerator - Batch 3, Germany, June-September 2022
- Selected as one of six “Impact Startups to Watch” in global sleep sector by Impact Shaker, July 2022
- Featured at 28th APEC SME Ministerial Meeting, Thailand, September 2022
- Winner (Social and Environmental Contribution), National Innovation Award 2022, Thailand’s most prestigious innovation award
- Member, Cambridge Cleantech, UK (since March 2023)

Our Founder & CEO is:

- One of four Finalists for the UK Alumni Entrepreneurial Award (Thailand) 2020
- On the inaugural Futurist Magazine’s The Future List, 2020
- A Tatler Asia Leader of Tomorrow (Generation T) 2020
- A Winner of the Business and Professional Women’s Association of Thailand - Bangkok under the Patronage of HM the Queen’s “Family Business Successor of the Year Award 2022”







Nophol Techaphangam
Founder & CEO

Family Business Successor of the Year Award
2022, Business and Professional Women's
Association of Thailand - Bangkok under the
Patronage of HM the Queen

Tatler Asia Leader of Tomorrow 2020

Futurist Magazine's 'The Future List' 2020

Finalist, UK Alumni Entrepreneurial Award
(Thailand) 2020

Model Business Person of the Year Award 2012
(Household and Office Goods Sector), Foundation
of Science and Technology Council of Thailand

Post Dip Economics
University of Cambridge

BSc Hons (1st) Physics with Theoretical Physics
University of Manchester



Parinda Sakdanaraseth
Communication Design
Director (Part-time)

Senior Product Designer
Calipsa

MRes Communication Design
Royal College of Art

BA Hons (1st) Graphic Design
Chulalongkorn University



Paphavee Sakdanaraseth
Web Interaction Design
Specialist (Part-time)

Senior Product Designer
Calipsa

MPhil Management
University of Cambridge

BA Hons (1st) Communication
Design, Chulalongkorn University



Kittikun Pensuk
Communication Director
(Part-time)

Communication Consultant

BA International Management
Cologne Business School



Chanida Pattanotai
Bond Market Specialist (Sell-Side)
(Project-based)

Certified Public Accountant &
SEC Financial Advisor (Thailand)

MSc Accounting
Chulalongkorn University

BA Accounting
Chulalongkorn University



Sitthi Teerakulchon
Bond Market Specialist (Buy-Side)
(Project-based)

Licensed Fund Manager

MSc Finance
Chulalongkorn University

BA Banking and Finance
Chulalongkorn University



Arinda Wijaya
Customer Relationship
Executive - Indonesia
(Part-time)

BSc in Communications,
Public Relations and
Business Administration
South Westphalia University of
Applied Sciences

OUR ADVISORS



Gavin Vongkusolkrit
Hospitality Industry

Director of the Board
The Erawan Group PCL.

Thailand's leading hotel investment company with over 6,100 rooms in 56 properties under 11 brands across all segments in Thailand's key destinations in its portfolio

Managing Director
Heritage Estates Co., Ltd.

A lifestyle investment company with Bangkok's most successful four-star boutique hotel, serviced offices, event spaces, F&B outlets and a 16,500-sq.m. office tower in Central Bangkok in its portfolio

Master of Business and Finance
 University of Sydney



Siripattra Kusuwan Na Ranong
Hospitality Industry

Deputy Managing Director
Pearl Group

Her family's business holding that includes a vast land bank, 212-room mid-tier business hotel, 184-room luxury resort, several major car dealerships and Phuket's only airline caterer in its portfolio.

Vice President
Thai Hotels Association
Southern Chapter

An association of 141 hotels and resorts in Southern Thailand

Former Consultant
McKinsey & Company SE Asia

PhD Biochemistry
 University of Oxford



Prasertsuk Patoonsittichai
Environmental Planning and Policy

Former Advisor to Board of Directors & Former Executive Director

Thailand Greenhouse Gas Management Organisation
(Public Organisation)



Chris Hampel
Furniture Industry

Former Group Director
Kian Group of Companies

Former Group Director
GARANT Group

Advisor to Board of Directors
Lifestyle and Design Products Association
Thai Chamber of Commerce

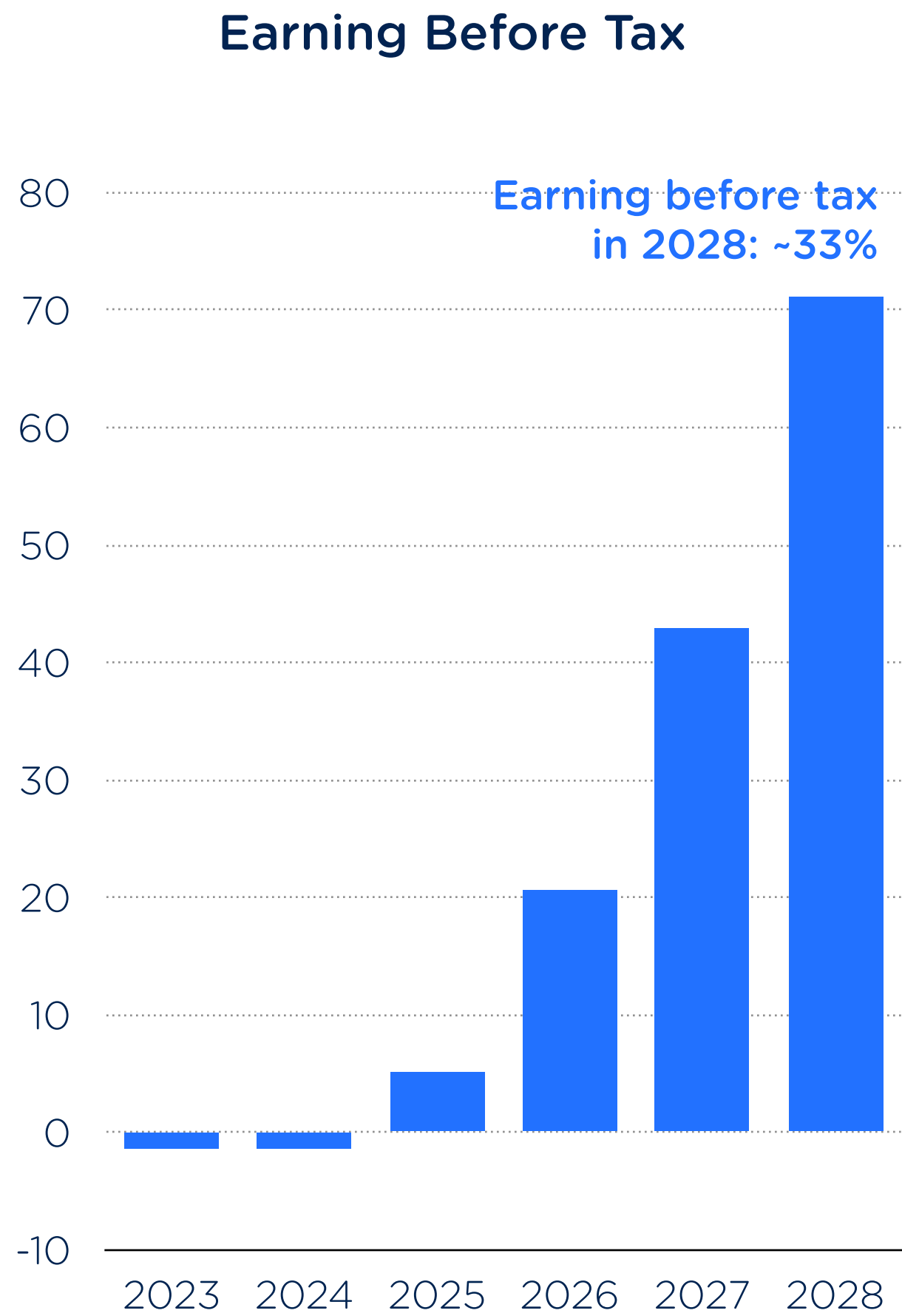
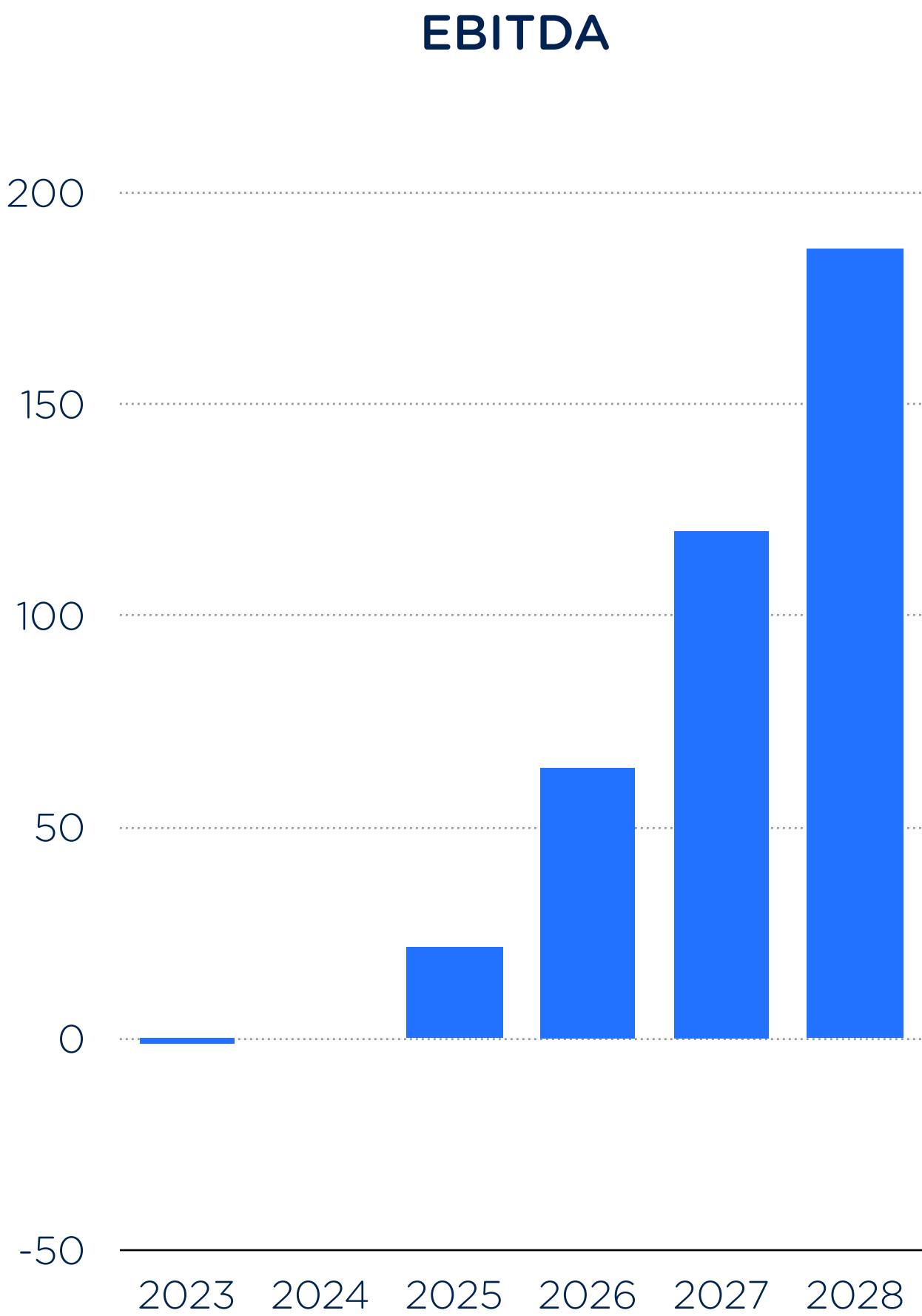
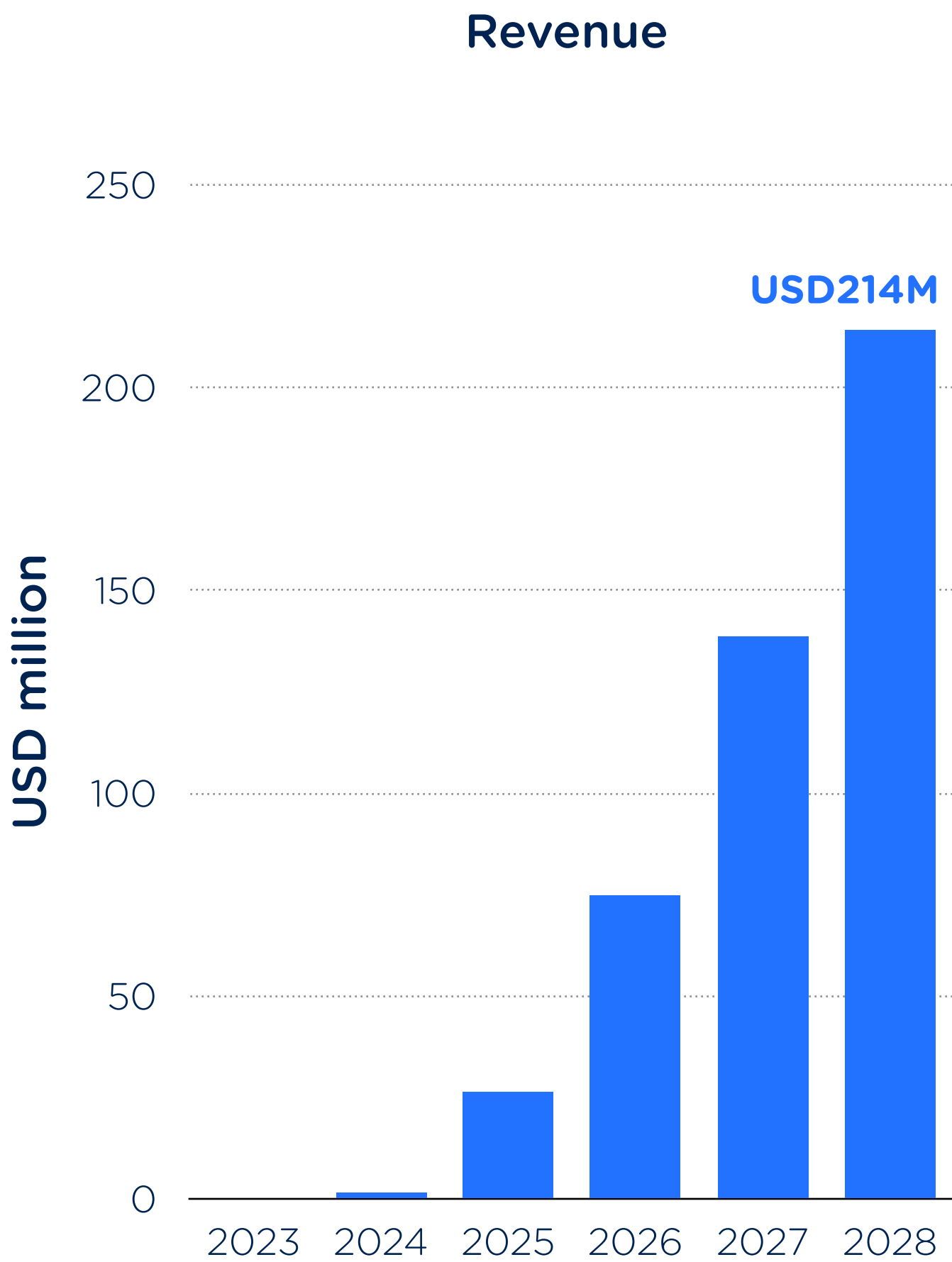


Paravi Wongchirachai
Communication Strategy

Co-Founder
Thailand Creative & Design Centre (TCDC)

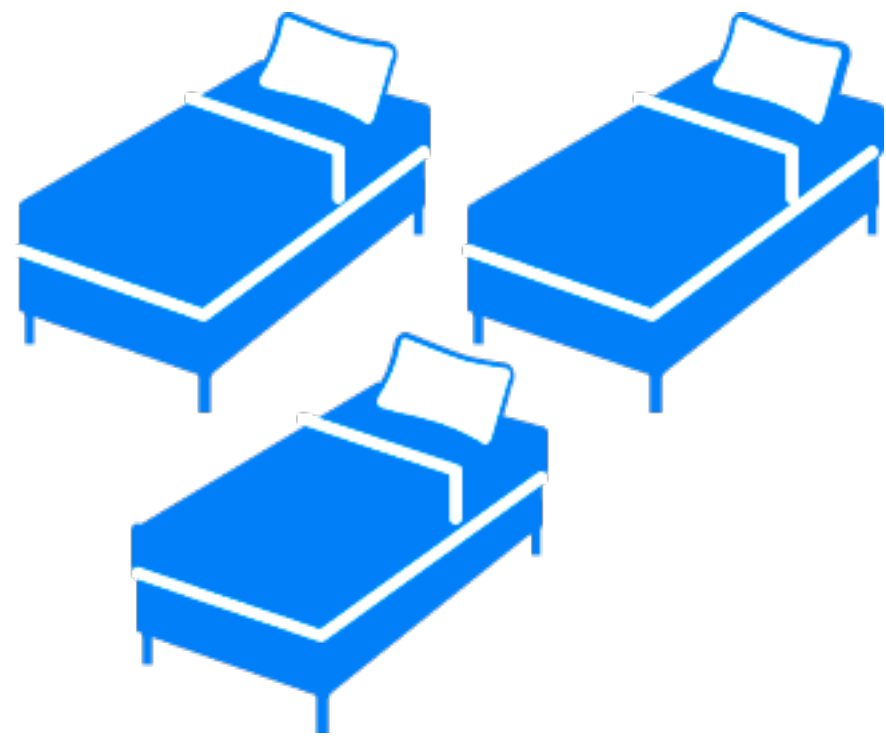
OUR FUTURE GROWTH: THAILAND & INDONESIA

2028 Cumulative no. of B2C orders: ~1.93M
Cumulative no. of B2B orders: ~58,530

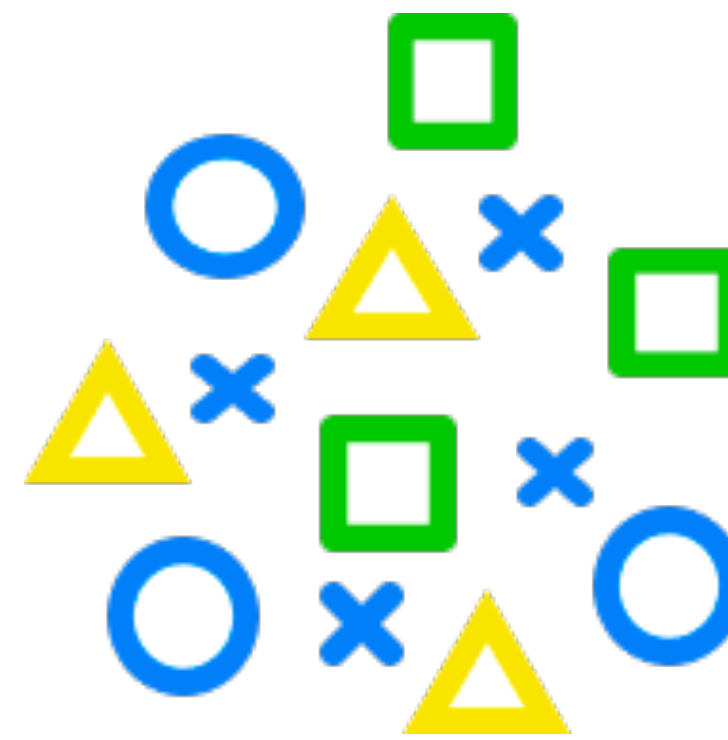


Forecast as of April 2, 2023 for Thailand’s B2C and B2B markets and Indonesia’s B2B market. USD:THB = 34.0 USD:IDR = 16,000

By the year 2028*:



**6.5M mattresses
to be saved from
becoming waste****



**285k tonnes
of materials to be
reclaimed and recycled**



**At least 265k tonnes
of CO2 emission saved*****

*Based on projection on Slide 28. **In standard size (107 x 198 x 24 cm.) equivalence

***Equivalent to removing at least 57,000 cars from the road for one whole year. | Reference: <https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-passenger-vehicle>

Mattresses are just the beginning.

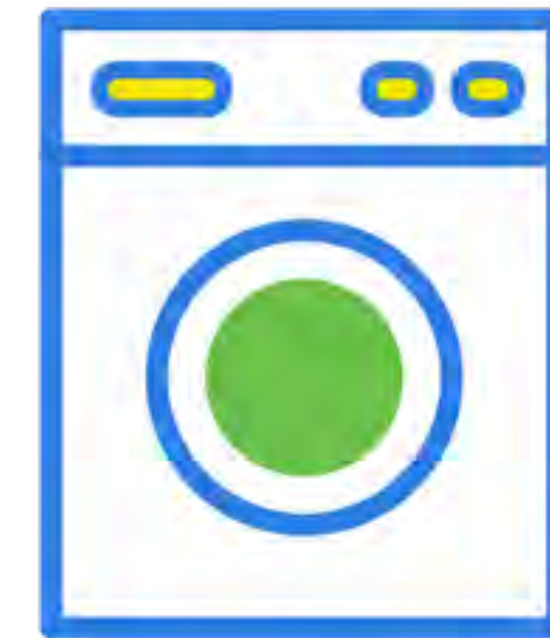
Next product categories:



Batteries



Solar Cells



Electronic Devices



free one month subscription if register and
confirm order by October 31 and refer to
“Hult Prize at Chula 2023”

For more information, contact:

Nophol Techaphangam
Founder & CEO
+66939149565
naps@nornnorn.com

